ITA0610-Machine Learning

LAB EXPERIMENTS

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Lab 5: Single-Layer Perceptron

Aim

To implement a **Single-Layer Perceptron** for binary classification.

Algorithm

1. Import the required libraries.
2. Create the input dataset and target output.
3. Initialize the Perceptron model.
4. Train the model using the given training data.
5. Predict the output for the input samples.
6. Display the predictions.

Program

```
# Lab 5: Single-Layer Perceptron

import numpy as np

# Input data
X = np.array([
    [0, 0],
    [0, 1],
    [1, 0],
    [1, 1]
])

# Target output for OR gate
y = np.array([0, 1, 1, 1])
```

```
# Initialize weights and bias
weights = np.zeros(2)
bias = 0
learning_rate = 0.1

# Train the perceptron
for epoch in range(10):
    for i in range(len(X)):
        linear_output = np.dot(X[i], weights) + bias

        prediction = 1 if linear_output >= 0 else 0
        error = y[i] - prediction
        weights += learning_rate * error * X[i]
        bias += learning_rate * error
# Test the perceptron
print("Predictions:")

for sample in X:
    output = np.dot(sample, weights) + bias
    prediction = 1 if output >= 0 else 0
    print(sample, "->", prediction)
```

Sample Output

```
Predictions:
[0 0] -> 0
[0 1] -> 1
[1 0] -> 1
[1 1] -> 1
```

Result

Thus, the **Single-Layer Perceptron was successfully implemented** and trained to perform binary classification for the OR gate.